

PRINCE SAMUEL KYEREMANTENG

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EDUCATION

Lund University, Sweden

Exchange Program; Master's in Computer Science
Erasmus+ Scholar; **About €6000 in grant.**

Lund, Sweden

Jan – Jun 2026

University of Ghana, Ghana

MPhil Computer Science; **CGPA 3.75/4.0**

Accra, Ghana

Oct 2024 – Jul 2026

Thesis: A Residual Hybrid Deep Learning Framework for Missing Data Imputation in IoT Sensor Data Using LSTM and Variational Autoencoders

University of Ghana, Ghana

Bachelor's in Computer Science; **CGPA 3.45/4.0**

Accra, Ghana

Oct 2019 – Oct 2023

Thesis/Project: SmartFlow: IoT-Based Smart Irrigation and Environmental Monitoring System

SKILLS SUMMARY

- **Languages:** Python, SQL, JavaScript, Dart, HTML, CSS, Tailwind
- **Frameworks:** Django, FastAPI, Flutter, PyTorch, TensorFlow, Pandas, ScikitLearn, React
- **Tools:** Git, Postman, PostgreSQL, PgAdmin, LLM, Agentic AI, RAG, Docker, REST API
- **Platforms:** GitHub, Jupyter Notebook, VSCode, Codex, Linux Servers, Overleaf
- **Soft Skills:** Project Management, People Management, Excellent Communication, Versatile

RESEARCH INTERESTS

- AI-Enabled Software Engineering (AI4SE, SE4AI, Agentic AI, etc.)
- Software Security and Secure Software Development
- Empirical Software Engineering/Development, Testing, and Optimization
- Artificial Intelligence (AI) and Machine Learning Systems Engineering (LLMs, Agent Orchestration)
- Secure learning for Wireless and IoT Systems, and Data Analytics

RESEARCH EXPERIENCE AND PUBLICATIONS

Research Experience

- Graduate Teaching and Research Assistant (TA/RA), University of Ghana 2023 – Present
- MPhil Thesis Researcher, University of Ghana Oct 2024 – Sep 2026

(1) Book Chapter: Kyeremanteng, P. S., et al (2025).

Federated Learning at the Edge: Enhancing Privacy and Efficiency in Distributed AI Systems

Publisher: [IntechOpen](#) || <https://www.intechopen.com/chapters/1230198> || **Co-author**

(2) A Review Manuscript: Kyeremanteng, P.S., et al.

A Systematic Review of Large Language Models, 2017 to 2026

Publisher: Preprints.org || **DOI:** [10.20944/preprints202607.0296.v1](https://doi.org/10.20944/preprints202607.0296.v1) || **First Author**

(3) Short Paper (Technical Report): Kyeremanteng, P.S., et al.

ClarifyFlow: A Clarification-Aware Agentic Pipeline for Code Generation.

LINK: <https://github.com/DevPrinceK/clarifyflow>

(4) A Review Manuscript: Kyeremanteng, P.S., et al.

Computational Intelligence for Missing Data Imputation: Techniques, Challenges, Innovations, and Applications - A Systematic Review

Publisher: International Journal of Computational Intelligence Systems || **Under Review** || **First Author**

(5) Article Manuscript: Kyeremanteng, P.S., et al.

Gap-Aware Prior-Guided Imputation with Conservative Residual LSTM-VAE Refinement for IoT Temperature Sensor Streams

Publisher: International Journal of Computational Intelligence Systems || **Under Review** || **First Author**

(6) Manuscript: Kyeremanteng, P. S., et al.

PoAD-Lite: A Lightweight, Activity-Aware Consensus Protocol for Scalable and Energy-Efficient IoT Blockchains

Publisher: IET Smart Cities || **Manuscript ID:** SMC-2025-10-0088.R2 || **Under Review**

(7) Manuscript: Kyeremanteng, P. S., et al.

Secure Sharding in PoAD: Enhancing Scalability Without Security Trade-offs

Publisher: IET Blockchain || **Manuscript ID:** BLC-2025-08-0170 || **Under Review**

CONFERENCES AND PRESENTATIONS

(1) Postgraduate Research Workshop, Computer Science Department – University of Ghana (2025)

Presentation: Missing Data Imputation Techniques: Applications, Challenges, and Innovations

Award: 3rd Best Paper Award

(2) Nyansapo AI Symposium on Sustainable Development

Organized by the Ghana Atomic Energy Commission and Université Paris-Saclay (2023)

Presentation: AI-Based predictive system for electricity outage forecasting.

WORK EXPERIENCE

Teaching and Research Assistant | [University of Ghana](#)

October 2023 – Present

- Conducted tutorials on machine learning, heuristic search, and knowledge representation.
- Assisted in research in the labs of two professors during my time as a research assistant.
- Assisted in teaching Artificial Intelligence, database management, and system design.

System Development Consultant | [SILEX Consulting Limited](#)

July 2025 - Present

- Led the development of ICCASA's Monitoring and Evaluation system.
- The system includes projects, indicators, outcomes, baselines, and forms for data collection.
- Automatic computations with custom formulae, with a comprehensive reporting system. The system includes tablet mobile apps for data collection by field agents.

Lead Software Engineer | [AfriLearnTech Ghana Limited](#)

September 2025 – Present

- Led the development of a national digital learning platform for Liberia's Ministry of Education.
- System includes LLM Agents for course content generation to augment localized content.
- System includes (synchronizable) offline boxes to be deployed in schools with low connectivity.

Software Developer | Iridis Labs**Sept 2023 – Jun 2025**

- Worked on a management software for the Electricity Regulatory Commission of Liberia.
- Recruited and onboarded 10 interns and conducted training sessions for them.
- Technologies used included: Flutter, PostgreSQL, React, TypeScript, and Ubuntu Servers.

Founder | PKay Software Consultancy**October 2023 – Present**

- Led the development of software systems, including web and mobile, for a wide range of clients.
- Systems include Admission Systems, Management Information Systems, M&E Dashboards, Cross-platform Mobile Applications, and Business Portals.
- Handled stakeholder engagements and the entire Software Development Life Cycle.

Software Developer | PayHub Ghana Limited**April 2021 - December 2025**

- Developed Payment Processing APIs for internal use and for clients.
- Developed fintech-grade dashboards for transaction monitoring and management.
- Reduced system latency through Redis-based caching and optimized session management.

PROJECTS**Transport and Hospitality Management Software | [Link](#)****July 2026**

- Built a holistic vehicle rental and guest house booking management system.
- System included payment integration, idempotent and atomic transaction processing.
- Comprehensive dashboard for managing and monitoring fleets and guest houses.
- System operates at a 99.9% availability/up time.

Projection Software | [Link](#)**April 2026**

- Built a desktop church projection platform using Tauri, React, and SQLite.
- Developed a real-time dual-window projection and display engine.
- Engineered offline content management for scriptures, hymns, and announcements.

AfriLearn Digital Learning Platform | [Link](#)**Oct 2025 - April 2026**

- Built an offline-first national digital learning platform for Liberia's Ministry of Education.
- System includes easy-to-deploy hardware 'boxes' deployable in low connectivity areas.
- Automatic updation and synchronization of educational content (videos, Audio, PDFs, etc.).
- Uses LLM to suggest and generate personalized lessons and assessments for learners.

Synther | [Link](#)**April 2026**

- A synthetic data generation tool for generating highly configurable datasets
- Implemented Celery workers and Redis for scalable background generation and live progress.
- Designed an intuitive wizard with datatype-aware fields, options, and downloads.
- Ensured data fidelity with numpy/Faker-based generators, and deterministic seeding.

Nrf24 Firewall FullStack and Embedded | [Link](#)**March – April 2026**

- Designed/implemented a lightweight software-defined firewall for an NRF24-based IP network.
- Built a full-stack system using FastAPI and React with a staged rule engine and web interface.
- Configured Linux TUN interfaces, routing, NAT, and integrated nftables for traffic filtering.
- Evaluated system performance using UDP throughput and latency measurements with iperf3.

ReviewMate | [Link](#)**November 2025**

- Designed and built a Word add-in with an LLM-powered review agent (TypeScript, Office.js).
- Designed prompt-driven review workflows to generate structured, actionable feedback as review comments in Word.
- Integrated OpenAI-compatible LLM pipeline with JSON outputs and resilient parsing.
- Added AI writing tools (trim, paraphrase, summaries) with secure API key handling.

Admissions Management System | [Link](#)**January 2024 – Present**

- Designed/developed an admissions management system for [GIMPA](#) (a tertiary institute in Ghana)
- The system includes a management portal for monitoring and management, with an automated and manual notification system.
- Over 2000 applications processed so far, and counting.
- Application processing time is improved by over 500 percent.

REFEREES**(1) Prof. Yaokumah Winfred**

Computer Science Department, University of Ghana

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Director, Africa Center for Development Impact (ACDI)

Director, SILEX Consulting Limited

Professor, Ghana Institute of Management and Public Administration

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